

AD 502 – Machine Learning

Pre-Reading Material – Lecture 18

Performance Measures in Classification

Why This Topic Matters

After building a classification model, we need to determine **how well the model is performing**. Performance measures help us compare the model's predictions with the actual class labels.

Two basic concepts are the **Confusion Matrix** and **Classification Accuracy**.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the purpose of a **confusion matrix**.
- Identify **True Positive, True Negative, False Positive, and False Negative**.
- Interpret a confusion matrix.
- Understand **classification accuracy**.
- Calculate accuracy from a confusion matrix.

1. Confusion Matrix

A **confusion matrix** is a table used to summarize the performance of a classification model by comparing the **actual classes** with the **predicted classes**.

For a binary classification problem, it contains four outcomes:

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Understanding the Four Outcomes

- **True Positive (TP):** The model predicts Positive and the actual class is Positive.
- **True Negative (TN):** The model predicts Negative and the actual class is Negative.
- **False Positive (FP):** The model predicts Positive, but the actual class is Negative.
- **False Negative (FN):** The model predicts Negative, but the actual class is Positive.

Key idea:

The word **True/False** tells whether the prediction is correct, while **Positive/Negative** tells which class was predicted.

2. Classification Accuracy

Classification accuracy measures the proportion of predictions that the model classified correctly.

It considers both:

- Correct Positive predictions
- Correct Negative predictions

A model with high accuracy makes a large proportion of correct predictions.

However, accuracy alone may not always provide a complete picture, especially when the classes are **imbalanced**.

3. Simple Example

Suppose a classifier makes **100 predictions**:

- 80 predictions are correct.
- 20 predictions are incorrect.

The classification accuracy is therefore **80%**.

Students should be able to relate this overall accuracy to the individual values in the confusion matrix.

Think Before the Lecture

1. What information does a confusion matrix provide?
2. What is the difference between **False Positive** and **False Negative**?
3. If a model predicts Positive and the actual class is Negative, which outcome is this?
4. Why might a model with high accuracy still need further evaluation?
5. If a classifier correctly predicts 90 out of 100 cases, what is its classification accuracy?

